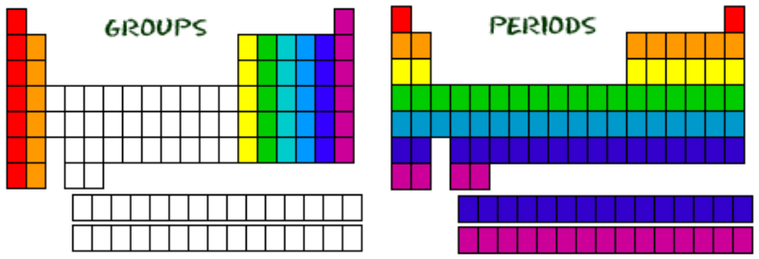
Knowledge Organiser
Year 8: Periodic Table – Using the PT

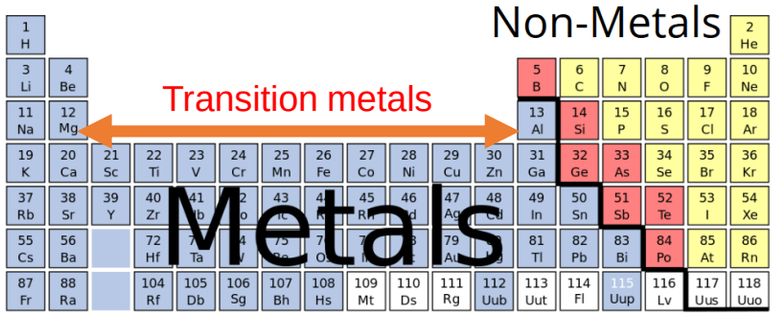
How is the periodic table organised?
- Elements are arranged in **groups (columns)** and **periods (rows)**
- Elements in the same group have s**imilar chemical properties** e.g. group 1 all react strongly with water
- Elements **increase in atomic mass** across a period

Diagram which shows the layout of **groups** and **periods** on the periodic table



Position of metals and non-metals?
- Metals are found to the left of the periodic table (below group 4)
- Non-metals are found to the right of the periodic table (group 4+)
- The block of elements between group 2 and 3 are called the **transition metals**
- The transition metals do not have a **group number** but all share some similar properties

Diagram which shows the position of metals, non-metals and transition metals on the PT



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Groups and properties?
- Each group has **shared chemical properties**
- This is because elements in the same group have the same number of **outer shell electrons**
- All that changes is **how reactive** the elements are e.g. lithium and sodium both fizz with water, but sodium also sparks

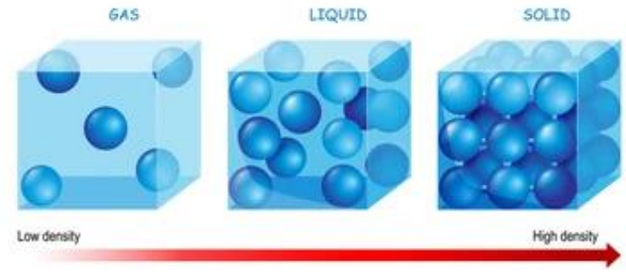
Group	Name	Properties	Trend in reactivity
1	Alkali metals	Soft, grey, low density, reactive metals	More reactive down the group
2	Alkaline earth metals	Strong, silvery, low density	More reactive down the group
7	Halogens	Low melting points, reactive, toxic	Less reactive down the group
8	Noble gases	Low melting points, very unreactive	Unreactive, do not form compounds

Metals vs. Non-metals?
- All elements can be assigned as either metals or non-metals
- These elements share most of their physical properties

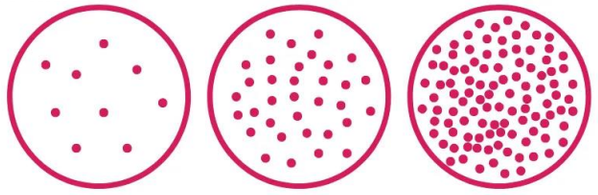
Properties	Metals	Non-metals
Density	High	Low
Melting points	High	Low
Conductivity	High	Low
Appearance	Silvery, shiny	White/colourless, dull
Malleability	Can be bent	Brittle
Strength	Strong	Weak

What is density?
- Density is the mass of an object divide by its volume
- A high density substance has particles which are packed more tightly together
- It is measured in grams per centimetre cubed (g/chm^3)

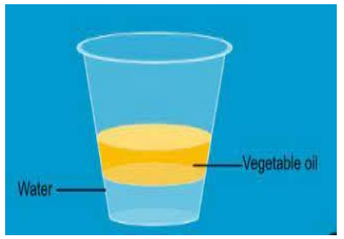
Particle diagram comparing the density of the three states of matter



Particle diagram comparing a low, medium and high density gas

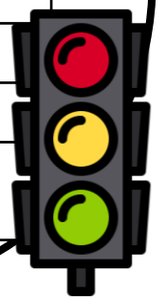


Why does oil float on water?
- Oil is less dense than water, its particles are less tightly packed together
- This means it experiences a weaker force of gravity than water



RAG Yourself!

Learning Outcomes;	RAG
Identify group and period from the periodic table	
State the names of groups 1, 2, 7 and 8	
Describe the properties of the transition metals	
Compare the properties of metals and non-metals	
Describe density for liquids and gases	
Explain why water sinks to the bottom of a beaker of oil	



Knowledge Organiser
Year 8: Periodic Table – Using the PT

How is the periodic table organised?

Diagram which shows the layout of **groups** and **periods** on the periodic table

Groups and properties?

Group	Name	Properties	Trend in reactivity
1			
2			
7			
8			

What is density?

Particle diagram comparing the density of the three states of matter

Particle diagram comparing a low, medium and high density gas

Position of metals and non-metals?

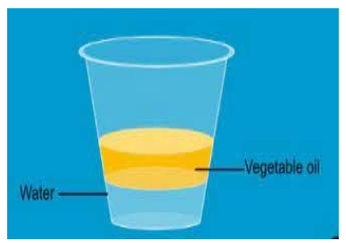
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Properties	Metals	Non-metals
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Melting points		
Conductivity		
Appearance		
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Why does oil float on water?



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